optimal technique

DP (Keywords: target, subsets, shortest distance, intervals)

↳ dynamic programming

① Recursion, backtrack.

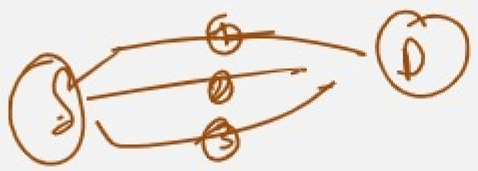
repetitive

① overlapping subproblems

② optimal substructure

[1, 2, 5, 6, 10]

caching



[4, 5, 6, 10, 2]

[1, 5, 6] [10, 2]

[1, 5] [6] [10] [2]

[1] [5]

Divide

Merging

0 1 1 2 5 8, 13, ----

Time complexity is in exponential.

$2^n \rightarrow O(n)$

f(5)

f(4) f(3)

f(3) f(2) f(2) f(1)

f(2) f(1) f(1) f(0) f(1) f(0)

f(0) f(1)

dp (caching)

0	∞	∞	∞	∞

↳ store the subproblems solution here

f(40)

f(39) f(38)

f(38) f(37) f(37) f(36)

Top-Down Approach (Recursive)

↳ Recursive

↳ cache the result of each subproblem.

↳ check if subproblems come again, we retrieve the data from cache.

Bottom-up approach

Iterative

↳ create a 1D or 2D arr.

↳ start by smallest, basic subproblem first.

↳ fill the 1D or 2D array with solution up to the top (main) problem.

0 1 1 2 3 5